CMPSC 448 Project 1: Who Wrote It?

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1. Abstraction

We worked on whether the source family of an AI-generated response can be recognized from its text. We built two classifiers, a convolutional neural network (CNN) and a bidirectional long short-term memory network (BiLSTM), to differentiate among GPT, Claude, and Gemini responses. We used a balanced 6,000-example subset of the public TXD-22 dataset and a prompt-aware train/validation/test split. In our test, we found that the CNN achieved 94.36% test accuracy and the BiLSTM achieved 92.26% on output-only classification. An input-only CNN achieved 33.41%, near the 33.33% chance baseline, which makes sense because the dataset is balanced, and nothing about the prompts tell us which model will answer it. Exploratory linguistic analysis showed differences in response length, sentence structure, and punctuation across the families.

2. Project Goal

We mainly focused on three LLM families, GPT, Claude, and Gemini, and tested whether their responses contain recognizable writing patterns. We also compared predictions made from the original prompt with predictions made from the generated response.

3. Dataset and Preparation

We used the public TXD-22 dataset and selected 2,000 examples each from GPT, Claude, and Gemini (6,000 total). Each example consists of 3 fields: The user prompt, the LLM response, and the model label. We split the data into 4,197 training, 899 validation, and 904 test examples, keeping the three classes balanced. Repeated prompts were grouped during splitting to reduce data leakage, where the text was tokenized and limited to 256 tokens per example.

4. Models and Training

The project uses both models. Our CNN turns words into 128-dimensional embeddings and uses filters of sizes 3, 4, and 5 to identify short text patterns. The bidirectional LSTM also uses 128-dimensional embeddings, but processes word sequences in both directions to capture context. Both models use dropout and predict one of three labels. We trained the CNN for 10 epochs and the BiLSTM for 8 epochs with batch size 32 and learning rate 0.001, then evaluated the checkpoint with the best validation accuracy.

5. Research Question 1 — Identifying the Source LLM

Model: CNN — output only; Best epoch: 10; Validation accuracy: 94.10%; Test accuracy: 94.36%

Model: CNN — input only; Best epoch: 1; Validation accuracy: 33.48%; Test accuracy: 33.41%

The CNN output-only classifier achieved 94.36% test accuracy (853 correct predictions out of 904), substantially exceeding the 33.33% chance baseline for three balanced classes. The best checkpoint was at 10 epochs based on 94.10% validation accuracy. This supports that the conclusion that generated responses contain identifiable patterns associated with their source LLM family.

Model: CNN; Family: GPT; Precision: 0.95; Recall: 0.92; F1: 0.93

Model: CNN; Family: Claude; Precision: 0.97; Recall: 0.96; F1: 0.96

Model: CNN; Family: Gemini; Precision: 0.92; Recall: 0.95; F1: 0.94

For the CNN, Claude had the highest precision (0.97). The confusion matrix, with rows and columns ordered by GPT, Claude, Gemini, was [[277, 6, 18], [6, 289, 7], [10, 4, 287]]. GPT and Gemini were confused more frequently with each other than either was with Claude.

LSTM results. The BiLSTM reaches 92.26% accuracy and 0.922 macro-F1.

Table 1. BiLSTM per-class results (output only, test set).

Class	Precision	Recall	F1	Support
GPT	0.862	0.977	0.916	301
Claude	0.959	0.937	0.948	302
Gemini	0.959	0.854	0.903	301
Macro avg	**0.927**	**0.923**	**0.922**	**904**

Table 2. BiLSTM confusion matrix (rows = true, columns = predicted).

	GPT	Claude	Gemini
GPT	294	3	4
Claude	12	283	7
Gemini	35	9	257

Claude is identified most reliably (F1 = 0.948). The largest residual error is Gemini misclassified as GPT (35 of 301). The LSTM therefore recovers distinctive stylistic signals for all three families, with the GPT/Gemini boundary being the hardest.

Yes. Using only the response text, both a BiLSTM and a TextCNN can identify the generating LLM family with high accuracy (LSTM 92.26%; CNN 94.36%), far above the 33.33% chance baseline. This confirms that GPT, Claude, and Gemini leave recoverable stylistic fingerprints in their outputs, and that both convolutional and recurrent architectures can learn these fingerprints effectively. In both models Claude is the most distinctive family and GPT vs. Gemini is the hardest distinction.

6. Research Question 2 - Does the User's Prompt Help Identify the LLM?

To understand whether the user's prompt helps identify the LLM, we trained our CNN using three different inputs: the prompt alone, the generated response alone, and by combining both.

The input-only model achieved 33.41% accuracy, which is close to random guessing among three LLMs. On the other hand, the output-only model achieved 94.36%, while combining the prompt and response resulted in 93.25% accuracy.

These numbers imply that the generated responses contain much stronger clues about which LLM produced them than the prompts themselves. Adding the prompt slightly reduced accuracy by 1.11

percentage points, suggesting that it did not give any useful information in our experiment. Overall, the CNN was most effective when classifying LLMs using only their generated responses.

LSTM comparison. We trained the BiLSTM three times, changing only the text that was tokenized and fed to the network:

- **Input only:** the user prompt.
- **Output only:** the model's response.
- **Input + output:** prompt and response joined by a <SEP> token, with the prompt budgeted up to max_length // 4 = 64 tokens and the response filling the rest of the 256-token window. In practice TXD-22 prompts are very short (median 8 tokens), so the response typically keeps ~247 of its tokens in this mode.

Vocabulary, architecture, optimizer, and random seed were held constant.

Table 3. LSTM comparison across input features (test set).

Features	Accuracy	Macro-F1	GPT F1	Claude F1	Gemini F1
Input only	33.41%	0.265	0.371	0.425	0.000
Output only	92.26%	0.922	0.916	0.948	0.903
Input + output	90.60%	0.906	0.894	0.937	0.888

On TXD-22, the user's prompt does not help identify the LLM, and including it slightly hurts. Three findings support this:

1. **Input-only performance is at chance.** 33.41% accuracy on a balanced 3-way task is random guessing, both models hit this. For the LSTM, the model collapses to predicting mostly GPT or Claude and never Gemini (Gemini F1 = 0), so the prompt carries essentially no evidence about which LLM answered.
2. **Input + output underperforms output alone** (LSTM: 90.60% vs. 92.26%; CNN: 93.25% vs. 94.36%). On TXD-22 the prompt is only ~8 tokens on average, so the response loses only ~9 of its 256 token slots in this configuration. The gap is therefore unlikely to come from truncation alone, a more plausible interpretation is that the uninformative prompt tokens act as a mild distractor for the model.
3. **Why this happens.** TXD-22 was built by sending the same prompts to every LLM, so each prompt appears with GPT, Claude, and Gemini responses. The prompt therefore carries no information about the label by construction. In a dataset where different prompts go to different models (e.g., LMSYS Chatbot Arena), this result might not hold.

7. Research Question 3 -Cross-Task Generalization

Setup. The RQ3 prompt asks whether an LLM classifier trained on one type of task can still identify LLMs on another type. TXD-22, however, consists entirely of factual question-answering prompts; there are no coding, creative-writing, or math tasks to swap between. The feasible generalization axis for this dataset is therefore the topic domain of the question. We partitioned the 2,034 unique prompts into two broad domains using keyword matching:

- **STEM:** prompts containing science, technology, engineering, or math keywords (e.g., nuclear, software, enzyme, voltage, probability, algorithm). 329 unique prompts; ~985 (prompt, response) examples balanced across GPT, Claude, and Gemini.
- **NON-STEM:** prompts containing social-science, humanities, media, legal, or arts keywords (e.g., gender, criminology, globalization, media, democracy). 496 unique prompts; ~1,461 (prompt, response) examples balanced across the three LLMs.

Prompts that matched both lists or neither were excluded from this experiment to avoid contamination.

We then trained a single BiLSTM (same architecture, optimizer, max length, and seed as the main experiments) on STEM responses only, with an 80 / 10 / 10 prompt-grouped split, giving 730 train / 90 validation / 96 held-out STEM test examples. The trained model was then evaluated on both the held-out STEM test set (in-domain) and the full 1,443-example NON-STEM set (cross-domain), without any fine-tuning.

Results. The BiLSTM trained on STEM reached a best validation accuracy of 85.56% at epoch 8.

Table 4. BiLSTM cross-domain evaluation (output only, trained on STEM responses).

Test set	n	Accuracy	Macro-F1	GPT F1	Claude F1	Gemini F1
STEM (in-domain)	96	82.29%	0.820	0.717	0.952	0.789
NON-STEM (cross-domain)	1,443	77.34%	0.765	0.679	0.858	0.759

Table 5a. BiLSTM confusion matrix, STEM in-domain test (n = 96; rows = true, columns = predicted).

	GPT	Claude	Gemini
GPT	19	0	13

	GPT	Claude	Gemini
Claude	1	30	1
Gemini	1	1	30

Table 5b. BiLSTM confusion matrix, NON-STEM cross-domain test (n = 1,443; rows = true, columns = predicted).

	GPT	Claude	Gemini
GPT	267	68	146
Claude	9	455	17
Gemini	30	57	394

Observations.

1. **Accuracy drops only ~5 percentage points out of domain** (82.29% → 77.34%). This is a modest, not catastrophic, decrease. It suggests the BiLSTM learned a representation that mostly captures LLM-stylistic signal rather than STEM-topic vocabulary.
2. **Claude generalizes best.** Claude F1 remains the highest in both settings (0.952 in-domain, 0.858 cross-domain, a 9.4-point drop), consistent with the main RQ1 finding that Claude's responses are the most distinctive.
3. **GPT drops only 3.8 F1 points and Gemini only 3.0.** Both start lower than Claude, so there is less performance to lose. The main cross-domain error is still between GPT and Gemini (146 GPT responses predicted as Gemini), which mirrors the error channel seen in the main RQ1 experiment.
4. **Caveat on the in-domain comparison.** The in-domain STEM test set is small (n = 96, 32 per class), so the 82.29% figure has non-trivial variance, while the cross-domain figure (n = 1,443) is much tighter. The direction of the gap is reliable even if its exact size is noisy.

7.2 Cross-task: Factual + Math → Coding (CNN)

TXD-22 does contain a small subset of task-specific prompts. We applied a regex-based taxonomy to all 2,034 unique prompts, labeling each as one of coding, math, creative, factual, or

unclassified. After filtering, we trained a CNN on all factual + math examples and evaluated it on held-out coding prompts, which the model never saw during training.

Table 6. CNN cross-task evaluation (output only, trained on factual + math).

Test set	n	Accuracy	Macro-F1	GPT F1	Claude F1	Gemini F1
Coding (cross-task)	321	87.85%	0.880	0.877	0.901	0.861

Table 7. CNN cross-task confusion matrix (coding test set; rows = true, columns = predicted).

	GPT	Claude	Gemini
GPT	89	3	15
Claude	3	91	13
Gemini	4	1	102

The CNN reached 87.85% test accuracy on coding, with best validation accuracy 92.90% during training on factual + math. This is only slightly below the main CNN output-only accuracy on random Q&A (94.36%), despite the test being on an entirely unseen task type. Claude is again the easiest to identify (F1 = 0.901), and the main residual confusion is between GPT and Gemini, the same pattern that appears in RQ1.

The two experiments agree: LLM fingerprints largely generalize both across topic domains (BiLSTM, STEM → NON-STEM, 77.3%) and across task types (CNN, factual + math → coding, 87.85%). In both cases accuracy stays well above the 33.3% chance baseline, and the per-model ordering (Claude easiest, GPT ↔ Gemini hardest) is preserved. Together these results suggest the classifiers are picking up relatively general stylistic signal rather than topic- or task-specific vocabulary. Both experiments remain limited to one dataset (TXD-22); a stronger test would require a corpus like LMSYS Chatbot Arena with substantially more task diversity.

8. Research Question 4 - Linguistic Fingerprint Analysis

We computed descriptive features over the 4,197 training responses to explore differences among the three families. The script reported word counts, sentence counts, mean sentence length, punctuation rates, and frequently occurring words.

LLM: GPT; Mean words: 247.554; Mean sentences: 10.800; Mean sentence length: 23.481; Commas / 100 words: 10.151

LLM: Claude; Mean words: 300.925; Mean sentences: 7.886; Mean sentence length: 40.031; Commas / 100 words: 9.052

LLM: Gemini; Mean words: 267.834; Mean sentences: 11.048; Mean sentence length: 29.656; Commas / 100 words: 9.541

Based on these results, Claude produced the longest responses on average and had far longer estimated sentences, while GPT responses were shorter with more sentences. Gemini was in between in word count and mean sentence length. GPT also had the highest comma rate among the three groups. These patterns are consistent with the corresponding stylistic variation, but they do not establish which features the neural networks have used. Feature ablations or explanation methods would be required to demonstrate causal contribution.

The frequent-word lists overlapped quite considerably. Majority of the examples included “such,” “their,” “while,” “through,” and “social,” suggesting that common content vocabulary alone is unlikely to explain all of the observed separation. Some patterns may also reflect the subject matter and formatting of the source dataset rather than stable characteristics of the model families.

9. Challenges, Limitations, and Lessons Learned

Model compatibility. During collaborative development, an older BiLSTM checkpoint used a 256-dimensional classifier input, while the updated implementation expected 512 dimensions. This prevented checkpoint loading until the model was retrained. We learned that checkpoints must be tied to the exact architecture and configuration used during training.

Overfitting and model selection. BiLSTM training accuracy reached 99.57% by epoch 8 while validation accuracy peaked at 93.21%, so there was a visible gap between training and validation performance. We saved the best validation checkpoint rather than the final epoch, which captured the model before this gap grew further. CNN validation accuracy continued improving through epoch 10."

Reproducibility. We verified the repository from a separate fresh clone, installed dependencies, regenerated the dataset partitions, trained both output-only models, evaluated them, and reran linguistic analysis. We noticed that the fresh CNN result exactly reproduced 94.36% test accuracy. We also learned to exclude raw datasets, generated CSVs, and large checkpoints from version control while documenting the external dataset source, since every machine gives subtly different results.

Scope and validity. The benchmark includes only three selected model families and one dataset. High within-dataset accuracy should not be interpreted as reliable identification of AI text. Prompt-aware splitting helps prevent exact-prompt overlap but does not prove transfer across topics, domains, model versions, or collection procedures.

10. Conclusion

Both required models successfully classified the source family of generated responses, with 94.36% CNN and 92.26% BiLSTM held-out test accuracy. The input-only CNN result suggests that attribution primarily relies on properties of generated responses in this benchmark. Descriptive linguistic differences offer plausible explanations, especially response and sentence length, but further controlled and in-depth experiments are required to establish robust, transferable LLM fingerprints.